

Home Assignment – Computer Vision Dataset Visualization Tool

You are joining a team that works closely with computer vision researchers. Researchers frequently work with large datasets and need tools that help them understand, inspect, and explore their data.

Your task is to build a web-based visualization tool for the Flickr8k dataset:
<https://huggingface.co/datasets/jxie/flickr8k/tree/main/data>

Requirements

- A React frontend
- A Python backend
- A visualization experience that helps users explore and understand the dataset

The application should allow users to browse dataset samples and inspect individual examples in more detail.

Beyond that, use your judgment to decide what capabilities would be most useful for a computer vision researcher working with this dataset.

Runtime Constraints

The entire solution must run locally on a single developer machine.

Please do not rely on cloud services, managed databases, external vector databases, hosted search services, or paid APIs.

The application should be runnable using standard open-source tools and libraries. All data required by the application should be stored and processed locally.

You may use local databases (e.g. SQLite) and open-source models or libraries that can run on a standard laptop.

We should be able to clone the repository, follow the setup instructions, and run the entire system locally.

Please keep the scope reasonable. We prefer thoughtful choices more than a large number of features.

Deliverables

- Source code
- Instructions for running the application (markdown file)

Evaluation Criteria

- Is the product useful for computer vision researchers?
- User experience and usability
- Frontend and backend design
- Code quality: is it generic, modular, and easy to expand and build on?

Contact

We intentionally leave many details unspecified, however, if you feel more clarification is needed, do not hesitate to contact us.

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Good Luck!